

Madhumithra M

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Professional Summary

Final-year B.Tech Computer Science (Information Security) student with strong Java, OOP, and Data Structures & Algorithms fundamentals. Built AI-powered and full-stack applications using Python, Flask, React, Next.js, and Firebase.

Education

Vellore Institute of Technology, Vellore, B.Tech Computer Science and Engineering (Information Security) Aug 2023 – May 2027

CGPA: 9.38 / 10.0

Government Girls Higher Secondary School, Pattukkottai

Mar 2023

Higher Secondary Education – 95.83%

Projects

AI PrepZone – AI-Powered Smart Learning Platform [\[GitHub\]](#)

- Built an AI-powered exam preparation platform using Gemini API and Flask for quiz generation and question paper analysis.
- Integrated Dialogflow chatbot and analyzed previous question papers to identify high-frequency topics.
- Technologies:** Python, Flask, Gemini API, Dialogflow, Pandas, HTML, CSS, JavaScript

Collaborative Sheets – Real-Time Spreadsheet Collaboration [\[Live Demo\]](#)

- Developed a real-time collaborative spreadsheet with live multi-user editing using Firebase.
- Implemented a custom formula engine, authentication, CSV export, and real-time synchronization.
- Technologies:** Next.js 14, React, TypeScript, Firebase, Tailwind CSS

VIT Scheduler – Classroom Allocation System [\[GitHub\]](#)

- Developed an Android classroom allocation system with Faculty and HOD role-based authentication.
- Enabled HODs to upload Excel timetables and manage classroom allocation, while Faculty could view assigned rooms and book halls/auditoriums..
- Technologies:** Kotlin, Android Studio, Room Database, SQLite

Patents

Published Indian Patent Application: System for Risk-Based Stateful Control of Artificial Intelligence Agent Interactions [\[Patent\]](#) 2026

- Publication No.: IN202641068105 A1.
- Designed a stateful, risk-based security framework integrating adaptive risk scoring, context management, and attack memory to protect AI agents and LLMs from advanced prompt-based attacks.

Certification

IBM Cybersecurity Analyst Professional Certificate [\[Certificate\]](#)

Completed IBM Cybersecurity Analyst training covering cybersecurity fundamentals, SOC, SIEM, SOAR, and incident management.

Technical Skills

Languages: Java

Core CS: Object-Oriented Programming (OOP), Data Structures & Algorithms

Web: HTML, CSS, JavaScript, React, Next.js

Databases: SQL, Firebase

Tools: Git, GitHub, VS Code

Achievements

VIT STARS Scholarship: Recipient of a 100% VIT STARS tuition scholarship for academic excellence.

Academic Excellence: School Topper in both 10th and 12th Grade Board Examinations.